

Investigative surgery

Effects of adenosine triphosphate administration in irreversible hemorrhagic shock

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It may seem strange that adenosine triphosphate (A.T.P.), once implicated as one of the depressor elements in traumatic shock,² is effective in the treatment of experimental shock in dogs. A.T.P. is involved widely in metabolism of the cell. It has an intimate role in carbohydrate metabolism and plays a part in tissue respiration.¹⁵ It supplies energy for these and many other intracellular reactions. It is the immediate source of energy for striated muscle, including the heart muscle contractions.¹² It is thought to unite with actomyosin to form the contractile substance, actomyosin A.T.P. complex.¹⁷ For further information about the biologic activity of A.T.P., the reader is referred to the monograph of N. Green,¹⁰ Guyton's *Textbook of Medical Physiology*,¹¹ and the papers of the Japanese A.T.P. Research Council.²² Since the initial pathophysiology in hemorrhagic shock is hypoxia at the cellular level, this report is concerned with the effect of A.T.P. on the survival of dogs subjected to irreversible hemorrhagic shock.

METHOD

Irreversible hemorrhagic shock was produced experimentally in 60 dogs using the elevated reservoir technique.^{20, 21} The animals were divided into 4 groups. The control groups consisted of 13 and 10 dogs averaging 13.9 and 18.0 kilograms in weight. They were anesthetized with intravenous pentobarbital sodium (25 mg. per kilogram of body weight) and a cuffed endotracheal tube was inserted. Aqueous heparin was injected intravenously (3 mg. per kilogram). The left femoral artery and vein were cannulated, using aseptic technique, and both cannulas were connected with a calibrated, sterile, siliconized glass bottle serving as a mobile reservoir. The arterial cannula had a side arm connected to a pressure transducer. The end-tidal CO₂, electrocardiogram (ECG), electroencephalogram (EEG), and systemic blood pressure were recorded continuously on a Grass polygraph recorder. Bleeding occurred through the arterial cannula into the reservoir, which was placed initially at the heart level of the dog. When the mean blood pressure had fallen to 35 mm. Hg, the reservoir level was adjusted to maintain that pressure throughout the experiment. A state of shock at 35 mm. Hg mean arterial pressure was maintained for 5 hours, or until 20 percent of the shed blood had been taken up spontaneously. These criteria are generally accepted as

Presented at the American Physiological Society Meeting, August, 1963.

Submitted for publication Oct. 14, 1963. Resubmitted Jan. 3, 1964.

*Fulbright Exchange Scholar on Travel Grant from Faculty of Medicine, Cairo University.

**Recipient of United States Public Health Service Career Research Development Grant 5-K3-NB-17, 651.

indicative of the irreversibility of shock in the dog.^{8, 9, 21} The rest of the blood was then retransfused through the intravenous cannula. Polybrene, diluted in 200 ml. of 5 percent dextrose, was given intravenously after completion of the retransfusion in amounts antagonistic to half the dose of heparin. The cannulas were then removed with ligation of the femoral artery and vein, and the dog returned to its cage.

In the second group of 10 dogs, averaging 16.6 kilograms in weight, the same technique as described for the controls was employed, but with the addition of 100 mg. A.T.P.* to the reservoir immediately after the initial bleeding when the blood pressure had reached 35 mm. Hg. Two hours after the retransfusion of blood, the dog was given another 100 mg. A.T.P. intramuscularly.

The third group, consisting of 17 dogs averaging 16.8 kilograms in weight, was subjected to the same procedure, but premedicated with an intramuscular injection of 100 mg. A.T.P. 1 hour before the start of the experiment. An additional 100 mg. A.T.P. was added to the reservoir immediately after the initial bleeding when the blood pressure was 35 mm. Hg.

A fourth group of 20 dogs was carefully selected. These animals were divided into 2 groups of 10 dogs. Each dog in Group 4A was meticulously paired with a dog in Group 4B by age, sex, weight, and predominant breed. Dog No. 1 in Group 4A thus is the paired control for Dog No. 1 in Group 4B, and similarly for the remainder of the animals. Splenectomy was performed one month prior to the shock experiment on all dogs. The A.T.P.-treated dogs received 6.5 mg. per kilogram of A.T.P. intramuscularly 1 hour before the shock experiment and another 6.5 mg. per kilogram in the reservoir immediately after the initial bleeding when the blood pressure was 35 mm. Hg. Splenectomy and the shock experiment were carried out simultaneously on each treated animal and his control. A.T.P. was administered,

after random selection, to one of the two animals by another person without informing the shock team of the animal selected. The animals were bled as rapidly as possible until the blood pressure had dropped to 35 mm. Hg. The first 4 pairs of dogs were maintained at a blood pressure of 35 mm. Hg for 5 hours, or until 20 percent of the shed blood was taken up. The period of shock was extended in the next 3 pairs of dogs to 6 hours, or until 30 percent of the shed blood was taken up. In the last 3 pairs of dogs the period of shock was reduced to 5½ hours, or until 25 percent of the shed blood was taken up spontaneously.

Rectal temperature was measured at the start of the procedure and at the end. The pH of arterial blood was measured in some animals of the third group at the start of bleeding, before transfusion, and two hours after retransfusion. All dogs were sacrificed 7 days after the experiment.

RESULTS

In all groups a number of dogs died early in the experiment of cardiac arrest. These animals are listed in the tables, but excluded from the statistical analysis. It seemed reasonable to exclude these animals to more appropriately test the influence of a therapeutic agent which was administered in part after the initial bleeding. The results are shown in Tables I, II, III, IV, and V and summarized in Table VI. In the first control group of 13 dogs, 4 died of cardiac arrest early in the experiment (30.7 percent). Of the 9 remaining dogs, 7 died within 72 hours, and the remaining 2 survived, i.e., the survival rate was 22 percent. This corresponds to most of the previously reported mortality data on irreversible shock in the dog.^{1, 7} The "bleeding time," or time elapsed between onset of hemorrhage and drop of blood pressure to 35 mm. Hg, of this group averaged 17.8 minutes. The amount of blood lost averaged 47.2 ml. per kilogram. The shock time averaged 207 minutes. The spontaneous uptake of blood from the reservoir during the shock period was 19.2 percent of the total amount shed.

*Adenosine 5-triphosphate disodium (3H₂O crystalline). Sigma Chemical Company, St. Louis, Mo.

Table I. Control Group 1

Dog No.	Weight (kg.)	Start- ing tem- pera- ture (° C.)	Initial blood pres- sure (mm. Hg)	Total amount bled (ml.)	Propor- tional amount bled (ml. per kilo-gram)	Bleed- ing time (min.)	Shock time (min.)	Uptake (%)	Tem- pera- ture at end of ex- peri- ment (° C.)	Blood pres- sure after rein- fusion	Blood pres- sure 1 hour after rein- fusion	Sur- vival at 1 week
1	Cardiac arrest											
2	Cardiac arrest											
3	Cardiac arrest											
4	11.3	38	120	550	48.6	10	196	20	36	120	110	Died
5	Cardiac arrest											
6	20.9	39	140	970	46.4	15	106	20	37	130	130	Died
7	10	38	90	400	40	30	275	25	36	90	90	Lived
8	11.6	39	120	600	51.7	30	166	12.6	37	110	110	Lived
9	15.4	38	120	800	51.7	20	155	20	36	120	120	Died
10	12	38	120	600	50	14	151	20	36	120	120	Died
11	12.5	38	110	550	44	13	239	20	36	90	100	Died
12	16.8	38	100	1000	59.5	10	275	20	36	100	90	Died
13	15	37	80	500	33.3	19	300	16	34	75	75	Died
Aver- age	13.9		111	663	47.2	17.8	207	19.2				

Table II. Postshock A.T.P., Group 2

Dog No.	Weight (kg.)	Start- ing tem- pera- ture (° C.)	Initial blood pres- sure (mm. Hg)	Total amount bled (ml.)	Propor- tional amount bled (ml. per kilo-gram)	Bleed- ing time (min.)	Shock time (min.)	Uptake (%)	Tem- pera- ture at end of ex- peri- ment (° C.)	Blood pres- sure after rein- fusion	Blood pres- sure 1 hour after rein- fusion	Sur- vival at 1 week
1	12.7	37	125	650	51	15	354	38.5	35	95	95	Lived
2	16.1	38	100	750	46.5	12	285	40	39	100	85	Died
3	20.3	37	120	1000	49.2	8	238	20	35	100	100	Died
4	Cardiac arrest											
5	15	39	120	750	50	13	193	20	37	120	120	Died
6	Cardiac arrest											
7	20.4	38	125	1100	54	11	143	10	36	120	120	Lived
8	Cardiac arrest											
9	15.6	37	110	600	38.4	11	273	25	35	100	100	Lived
10	25.4	38	110	1400	55.1	10	310	14.2	36	110	110	Lived
Aver- age	17.9		115.7	892.8	49.8	11.4	256	23.7				

In the second group of 10 dogs, which were treated with A.T.P. after the develop-
ment of irreversible shock, 3 died of cardiac
arrest during the experiment (30 percent).
Of the remaining 7, 4 survived and 3 died
within 72 hours of the experiment. This
constitutes a survival rate of 55 percent.
The bleeding time of this group averaged

11.4 minutes. The amount of bleeding aver-
aged 49.8 ml. per kilogram and the shock
time 256 minutes. The uptake of blood from
the reservoir during the shock period was
23.9 percent of that shed during the initial
hemorrhage.
In the third group of 17 dogs, all of which
were premedicated with A.T.P., 2 died of

Table III. Preshock A.T.P., Group 3

Dog No.	Weight (kg.)	Start-ing tem-pera-ture (° C.)	Initial blood pres-sure (mm. Hg)	Total amount bled (ml.)	Pro-por-tional amount bled (ml./ kg.)	Bleed-ing time (min.)	Shock time (min.)	Uptake (%)	Tem-pera-ture at end of experi-ment (° C.)	Blood pres-sure after rein-fusion	Blood pres-sure one hour after rein-fusion	Sur-vival at 1 week
1	16.1		100	700	44	15	300	15.1		90	90	Lived
2	17		140	900	53	37	300	11.1		100	100	Lived
3	16.8		120	750	44.6	32	263	20.1		110	110	Lived
4	16.5		120	900	54.8	30	294	22.2		100	100	Lived
5	20		140	1150	57.4	12	240	20		110	110	Lived
6	16.3		130	900	55	13	272	17.7		120	120	Lived
7	15.4		140	820	53.2	12	167	20.7		130	130	Lived
8	11.3		140	650	57.5	30	127	21.5		140	140	Died
9	12.2	39	120	700	57.3	15	180	20	37	110	110	Died
10	17.7	38	110	500	28.2	30	140	20	36	90	90	Lived
11	16.8	38	110	700	42.4	17	190	21.4	36	110	110	Lived
12	Cardiac arrest											
13	Cardiac arrest											
14	19.5	38	120	1400	71.8	30	270	15.7	35	120	120	Lived
15	17	38.5	120	850	50	20	300	22.3	35.5	120	120	Lived
16	21.3	37	120	1200	56.3	15	130	20	35	120	120	Lived
17	18.6	38	100	1150	61.8	10	220	20	36	100	110	Died
Aver-age	16.8		122	884.6	52.5	21	226	19.2				

Table IV. Paired control, Group 4A

Shock cri-teria	Dog No.	Weight (kg.)	Start-ing tem-pera-ture (° C.)	Initial blood pres-sure (mm. Hg)	Total amount bled (ml.)	Propor-tional amount bled (ml./ kg.)	Bleed-ing time (min.)	Shock time (min.)	Up-take (%)	Tem-pera-ture end of experi-ment (° C.)	Blood pres-sure after rein-fusion	Blood pres-sure 1 hour after rein-fusion	Sur-vival at 1 week
5 hours or 20 per cent uptake	1	14.1	38	110	550	39.7	8	170	20	36	135	120	Lived
	2	20.2	37	120	850	42	6	273	20	35	110	120	Died
	3	23.2	37	150	1100	47.4	13	273	20	35	110	110	Died
	4	14.5	36.5	135	800	55.1	9	256	20	35	120	110	Lived*
6 hours or 30 per cent uptake	5	20	36	120	950	47.5	5	360	25.2	36	120	120	Lived†
	6	15.4	37	110	850	55.2	6	360	29.4	36	100	100	Died
	7	Cardiac—Respiratory arrest											
5½ hours or 25 per cent uptake	8	19.3	37	110	950	49.2	9	276	25	35	90	90	Died
	9	15	36	135	800	53.3	9½	330	18.7	34	110	120	Died
	10	20	36.5	150	1200	60	11	142	25	33	140	100	Died
Aver-age		17.96	36.7	126.6	895	49.9	8.5	271.1	23.6				

*Died on 8th day.
†Died on 10th day.

Table V. A.T.P.-treated, Group 4B

Shock cri- teria	Dog No.	Weight (kg.)	Start- ing tem- pera- ture (° C.)	Initial blood pres- sure (mm. Hg)	Total amount bled (ml.)	Pro- por- tional amount bled (ml./ kg.)	Bleed- ing time (min.)	Shock time (min.)	Up- take (%)	Tem- pera- ture at end of ex- peri- ment (° C.)	Blood pres- sure after rein- fusion	Blood pres- sure 1 hour after rein- fusion	Sur- vival at 1 week
5 hours or 20 per cent uptake	1	Cardiac arrest											
	2	18	38	120	1000	55.5	6	300	11	36	110	120	Lived
	3	23.8	37	125	900	38.3	9	300	13.8	36	110	130	Lived
	4	15.2	36.5	125	775	51.0	6	300	14.2	35	100	120	Lived
6 hours or 30 per cent uptake	5	19.8	36	100	950	47.9	7	311	31.5	35	120	120	Lived
	6	17.3	37	130	900	52.0	6	360	27.7	36	110	110	Died
	7	17.7	38	130	1050	59.3	7	360	24.7	36	125	125	Died
5½ hours or 25 per cent uptake	8	22.7	37	120	1050	46.3	8	294	25	35	110	120	Lived
	9	15.7	36	130	850	54.4	7	330	5.8	33.5	130	135	Lived
	10	20	37	120	925	46.2	6	254	25	36	90	100	Lived
Aver- age		18.9	37.1	122	933	50.1	6.9	312	19.9				

cardiac arrest and were excluded. Of the remaining 15 dogs, only 3 died within 72 hours, and 12 survived for more than 1 week, i.e., the survival rate was 80 percent. The bleeding time of this group averaged 21 minutes. The amount of bleeding averaged 52.5 ml. per kilogram and the shock time 226 minutes. Uptake of blood from the reservoir during the shock period was 19.2 percent of the total amount shed.

In the fourth control group of 10 dogs, Group 4A, 1 dog died of cardiac arrest and

was, therefore, excluded. Of the remaining 9 dogs, 5 died within a period of 72 hours, 1 each died on the fifth, eighth, and tenth days, and another survived for an indefinite period. The survival rate at one week was 33 percent, and the bleeding time of this group was 8.5 minutes. The amount of bleeding averaged 49.9 ml. per kilogram and the shock time was 271 minutes. Uptake of blood from the reservoir during the shock period was 23.6 percent of the total amount shed.

Of the 10 dogs treated with A.T.P.,

Table VI. Summary

Group No.	Num- ber of dogs	Wt./ kg.	Initial blood pres- sure (mm. Hg)	Total amount bled (ml.)	Propor- tional amount bled (ml./ kg.)	Bleed- ing time (min.)	Shock time (min.)	Uptake (%)	Cardiac arrest		Mean sur- vival at 3 days (hours)	Sur- vival rate for 1 week (%)
									%	No.		
1	13	13.9	111	663	47.2	17.8	207	19.2	30.7	4	16.8	22
2	10	17.9	115	892	49.8	11.4	256	23.9	30.0	3	40.8	55
3	17	16.8	122	884	52.5	21.0	226	19.2	11.7	2	57.6	80
4A	10	18.0	126.6	895	49.9	8.5	271.1	23.6	10.0	1	51.5	33*
4B	10	18.9	122.0	933	50.1	6.9	312.0	19.9	10.0	1	64.7	76

*Two more control dogs died 8 and 10 days postshock.

Group 4B, 1 died of cardiac arrest and was excluded. Of the 9 remaining dogs, 2 died within 72 hours. Each of these 2 dogs was in the group of 3 with shock criteria extended to 6 hours, or 30 percent uptake. The over-all survival rate, however, was 76 percent at the end of 1 week, and no animals died after this period. The bleeding time of this group averaged 6.9 minutes. The amount of bleeding averaged 50.1 ml. per kilogram and the shock time was 312 minutes. Uptake of blood from the reservoir during the shock period was 19.9 percent of the total amount shed.

Thus, in summary, of the Group 4 dogs, the A.T.P.-treated animals had a much higher survival rate than their paired controls and seemed to bleed a little more rapidly with a longer shock period and less spontaneous uptake of shed blood.

Rectal temperature usually dropped 1° to 2° C. during the course of the experiments. Almost all dogs in all groups had diarrhea and melena.

DISCUSSION

These experiments suggest a beneficial effect of A.T.P. on irreversible shock under the conditions described. The survival rate 1 week after the experiment was increased from 22 and 33 percent in the control groups to 80 and 76 percent in the group treated with A.T.P. There may be a beneficial influence of A.T.P. administered shortly after a major bleeding with blood repletion, but this is equivocal because of the small number of dogs in Group 2. The results cannot be explained solely on the basis of energy supply because the doses given are not large enough for this purpose. The general biological effect of A.T.P., injected intravenously, is to lower the blood pressure by splanchnic vasodilatation with compensatory vasoconstriction in skeletal muscle, kidney, and spleen, similar to the response to irreversible shock.^{6, 10} This action is brief because A.T.P. is rapidly dephosphorylated. The shock-inducing and lethal doses of A.T.P.¹⁰ are 35 times greater than the therapeutic doses used by us. In the normal dog or the dog in hemorrhagic

shock, the therapeutic doses employed in these experiments, when given intramuscularly or by intravenous drip, have no toxic or hypotensive effects. A.T.P. determination in blood and body tissues during shock shows a marked quantitative decrease.¹⁶ Recently, it has been suggested that A.T.P. has a beneficial effect on myocardial ischemia.¹⁸ Apart from being a potent coronary vasodilator, it increases cardiac consumption of glucose, lactic acid, and pyruvic acid, and lowers the rate of O_2 consumption. Actually, it should be beneficial in the management of ischemia in other tissues; for example, it may protect the liver from ischemic necrosis.¹⁹ Its beneficial effect on anoxia of the brain has been reported recently.⁵

In reviewing briefly some of the mechanisms claimed to cause irreversibility in shock, the following theories have found widespread acceptance: (A) Central cardiac failure.⁴ (B) Cell damage due to anoxia.³ (C) Bacterial endotoxins.⁷ (D) Sludging and intravascular thrombosis.¹⁴ We feel that A.T.P. acts as a myocardial support; and protects the cell against anoxia by pathways yet undefined.

There is a balance between A.T.P. and A.D.P. (adenosine diphosphate) in the energy release from glucose. This energy release ceases when A.T.P. or A.D.P. is utilized more rapidly than it is formed. It seems reasonable, therefore, to speculate that the administration of A.T.P. before and early in the course of hemorrhagic shock might act as a "pump primer" in the restoration of this essential balance. If this speculation is correct, then there must also be an adequate energy substrate of glucose, fructose, and galactose. It is of interest that injured patients have a diabetic-type glucose tolerance curve the first day or two after shock, suggesting a failure of utilization of glucose. Also, they are more resistant to insulin. The magnitude of these findings was proportional to the severity of the injury.¹³ This might be due to a persistence of the deficit of A.T.P. in the heart and other organs during and following shock.²³ A great deal of work remains to be done in this general area as

it offers promise in the management of patients in shock and in the pretreatment of those patients going to the operating room where shock might be expected. These observations are presented in the hope that they may stimulate interest in the experimental study of these compounds, their substrate, and their metabolic pathways. It may be advisable to give A.T.P. intramuscularly as a premedication to patients undergoing operations in which considerable blood loss is anticipated.

SUMMARY AND CONCLUSIONS

Irreversible hemorrhagic shock was induced in 60 dogs divided into 4 groups. The groups of 13 and 10 dogs, serving as controls, had a survival rate of 22 and 33 percent. The second group of 10 dogs, treated with A.T.P. after irreversible shock, had a survival rate of 55 percent. The groups of 17 and 10 dogs, treated with A.T.P. 1 hour preceding and immediately following hemorrhage, had a survival rate of 80 and 76 percent. Administration of A.T.P. under these experimental conditions appears to be of value in the prophylaxis of shock, especially in the prevention of irreversibility. The beneficial effect of A.T.P. administration is markedly diminished after shock is manifest or has become irreversible. The exact dosage and the optimal rate, mode, and route of administration merit further investigation.

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